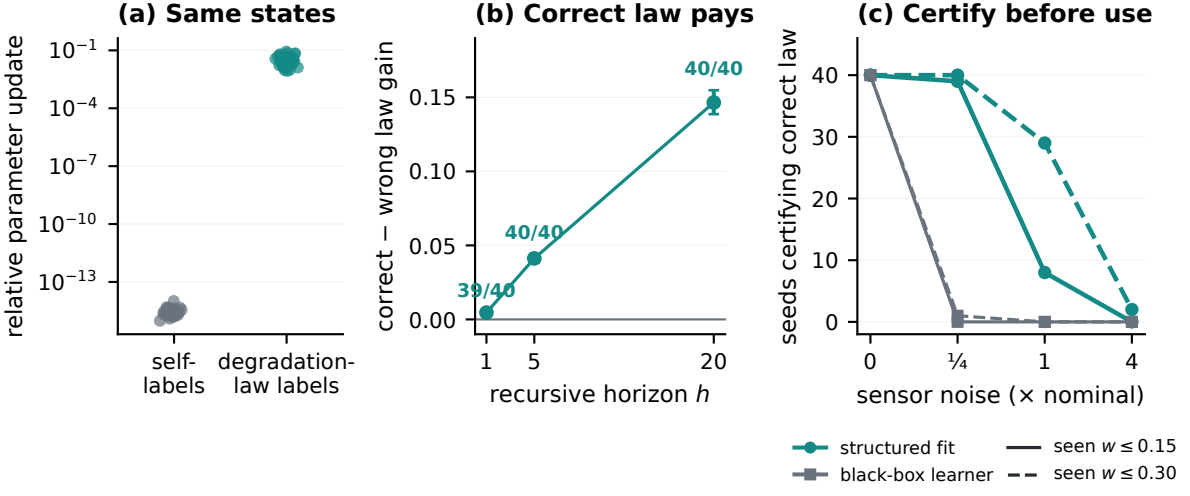
(a) Same states
relative parameter update
10^{-1}
10^{-4}
10^{-7}
10^{-10}
10^{-13}
self-labels
degradation-law labels
(b) Correct law pays
correct − wrong law gain
0.15
0.10
0.05
0.00
40/40
40/40
39/40
1
5
20
recursive horizon h
(c) Certify before use
seeds certifying correct law
40
30
20
10
0
0
¼
1
4
sensor noise (× nominal)
structured fit
black-box learner
seen $w \le 0.15$
seen $w \le 0.30$